

OMNIX

Decision Governance Infrastructure
for Automated Systems

"Preventing costly mistakes before they happen"

Eureka! GCC 2026 — Semifinalist Report

Startup: OMNIX

Founder & CEO: Harold Nunes

Stage: MVP (Live Product)

Pre-Seed Round: \$500,000 USD

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1. Executive Summary

Problem: High-stakes automated decision systems — from trading to supply chain to lending — lack institutional-grade governance infrastructure, leading to billions in preventable losses annually.

Solution: OMNIX is Decision Governance Infrastructure that validates every automated decision through 6 independent checkpoints before execution, with a fail-closed architecture that defaults to protecting capital.

Target Customer: B2B — Prop trading firms, trading platforms, regulated funds, and family offices in ADGM/DIFC ecosystem (200+ registered firms). Future verticals: supply chain operators, lending platforms, insurance underwriters, and robotics/autonomous physical systems (Year 3+).

Business Model: B2B Decision Governance Licensing (\$15K–\$35K/month per enterprise client) + per-validation pricing (\$0.01–\$0.05/call) for high-volume platforms. Secondary B2C SaaS (\$149–\$499/month).

Current Stage: MVP (Live Product) — Running 24/7 in production since November 2025. 670,000+ evaluation cycles completed, 16,000+ cryptographically signed governance receipts, 100% post-quantum cryptography coverage.

Key Traction: Eureka GCC 2026 Semifinalist. Live public verification system at omnibotgenesis-production.up.railway.app/verify. Interactive governance demos for credit/lending and insurance underwriting verticals. Institutional website live at www.omnixquantum.net.

Metric	Description	Context
670,000+	Evaluation Cycles	Governance engine operational 24/7
16,000+	PQC-Signed Receipts	100% Dilithium-3 coverage
98.5%	Capital Preserved	During BTC -7.37% volatility
<120ms	Decision Latency	Real-time governance validation
\$49.7B+	Combined TAM	Multi-vertical addressable market (6 verticals)

2. Company Overview

Vision

To become the global standard for decision governance infrastructure in automated systems — ensuring that every high-stakes automated decision is validated, traceable, and accountable before execution.

Business Description

OMNIX builds Decision Governance Infrastructure for automated systems. We provide a multi-layer pre-execution validation engine that governs decisions across domains where capital is at risk. Our 6-checkpoint architecture acts as an independent governance layer: it does not generate signals — it validates and governs them. The architecture is domain-agnostic, validated first in digital asset trading (the hardest domain due to 24/7 operation, high volatility, and zero circuit breakers), with expansion planned (Year 2–3+) into supply chain, credit/lending, insurance, energy trading, and robotics/autonomous systems.

Product Description

OMNIX Decision Governance Engine is a real-time, AI-powered validation infrastructure that sits between decision systems and execution. Every decision passes through 6 independent checkpoints: Monte Carlo Validation, Risk Management System Limits, Adaptive Coherence Gate, Edge Confirmation Window, Weighted Scoring, and Final Decision. The system operates on a fail-closed principle — if any checkpoint raises concern, the decision is blocked. Every decision (executed or blocked) is cryptographically signed with NIST-standardized post-quantum algorithms (Dilithium-3) and stored with a full audit trail.

Current Stage of Development

Component	Status
Decision Governance Engine	Live — Production 24/7 since Nov 2025
6-Checkpoint Validation	Live — All 6 checkpoints operational
Post-Quantum Cryptography	Live — Dilithium-3 + Kyber-768 since Nov 2025
Public Verification Server	Live — Public receipt verification endpoint
Investor Dashboard	Live — 19 widgets, real-time metrics
Institutional Website	Live — www.omnixquantum.net
Multi-Vertical Demos	Live — Credit/Lending + Insurance interactive demos
Shadow Portfolio Analysis	Live — 670,000+ counterfactual events
Enterprise API	In Development — Q2 2026 target

3. Problem & Market Opportunity

Problem Statement

Automated decision systems across industries — algorithmic trading, supply chain procurement, credit approvals, insurance underwriting, and robotics/autonomous physical systems — operate without adequate pre-execution governance. The consequences are severe:

- **\$68B+ lost annually** by traders alone due to emotional and undisciplined decision-making
- **95% of algorithmic systems** ask 'When should I act?' instead of 'When should I NOT act?'
- **Single-layer risk checks** fail during tail events (flash crashes, liquidity cascades)
- **No audit trail** for automated decisions, creating regulatory and compliance gaps
- **Institutional-grade governance** exists only inside hedge funds with \$100M+ AUM — inaccessible to the rest of the market

Existing Alternatives & Why They Are Inadequate

Alternative	Limitation	OMNIX Advantage
Retail Trading Bots	No decision governance, single risk check	6 independent checkpoint validation
Quant Fund Tools	Minimum \$10M+ capital, proprietary	Accessible infrastructure model
Manual Oversight	Too slow for real-time (<120ms required)	Automated validation in <120ms
Single-Domain Tools	Built for one industry only	Domain-agnostic governance engine
Traditional Risk Systems	React after the fact	Pre-execution validation (fail-closed)

Market Size

OMNIX addresses a multi-vertical market opportunity across decision governance for automated systems:

Vertical	Addressable Market	Timeline
Digital Asset Trading (Current)	\$18.8B algorithmic trading	Now (Validated)
Supply Chain Risk	\$3.2B supply chain analytics	Year 2–3
Credit / Lending Governance	\$7.4B credit risk management	Year 2–3
Insurance Underwriting	\$5.1B insurtech	Year 3+
Energy Trading	\$2.8B energy risk management	Year 3+
Robotics / Autonomous Systems	\$12.4B industrial robotics safety	Year 3+
Combined TAM	\$49.7B+	Progressive

Target Geography & Customer Segment

Primary Geography: ADGM/DIFC ecosystem (Abu Dhabi/Dubai) — 200+ registered prop trading firms, 100+ regulated funds, 50+ crypto-native platforms. **Expansion:** EU (MiCA compliance creating urgent demand), Asia-Pacific (Year 2–3).

Primary Segment: B2B Institutional (80% focus) — Prop trading firms, trading platforms, regulated funds, and family offices. **Secondary:** B2C SaaS for advanced independent traders (20%, post-enterprise validation).

Expected Market Adoption

Phase	Timeline	Target
Enterprise Pilots	Months 1–6	3 pilot agreements (prop firms/platforms)
Paid Licenses	Months 6–9	3 paying enterprise clients
Regional Scale	Months 9–12	\$50K+ MRR, 15–30 clients in MENA
Multi-Vertical	Year 2–3	Supply chain + lending governance pilots
Global Scale	Year 3+	EU MiCA, Asia expansion

4. Customer Validation & Insights

Discovery Method

Customer validation has been conducted through multiple channels: direct conversations with prop trading firm managers in the ADGM/DIFC ecosystem, participation in the Eureka GCC competition (semifinalist status achieved), institutional investor due diligence sessions, and live product demonstrations with the governance engine running in real-time production.

Key Insights Learned

- **Governance, not alpha:** Institutional buyers care about decision accountability and audit trails, not signal generation
- **Compliance urgency:** MiCA regulation (2025+) creates immediate demand for decision governance documentation
- **Risk-first positioning resonates:** 'Preventing costly mistakes' is a stronger value proposition than 'maximizing returns'
- **Fail-closed architecture is differentiating:** Institutions value systems that default to protect over systems that default to act
- **Multi-vertical potential recognized:** The same governance engine logic applies to credit, insurance, supply chain, and robotics/autonomous systems decisions

Willingness to Pay

Based on market research and competitive analysis of institutional risk infrastructure pricing:

Segment	Expected Price Range	Basis
Prop Trading Firms	\$10K–\$15K/month (pilot)	Comparable to existing risk tools; drawdown prevention ROI 2–8x
Trading Platforms	\$0.01–\$0.05/validation	Per-API-call model; volume-driven
Regulated Funds	\$25K–\$35K/month	Audit trail + compliance documentation value
B2C Advanced Traders	\$149–\$499/month	SaaS market benchmarks for institutional tools

Customer Testimonials

[Section reserved for 5 customer testimonials with name, role, contact details, and key feedback. Testimonials are being collected from pilot conversations and will be included in the final submission.]

#	Name	Role	Contact	Key Feedback
1	[To be added]	[Role]	[Email/Phone]	[Pain point, value perception, pricing sensitivity]

#	Name	Role	Contact	Key Feedback
2	[To be added]	[Role]	[Email/Phone]	[Pain point, value perception, pricing sensitivity]
3	[To be added]	[Role]	[Email/Phone]	[Pain point, value perception, pricing sensitivity]
4	[To be added]	[Role]	[Email/Phone]	[Pain point, value perception, pricing sensitivity]
5	[To be added]	[Role]	[Email/Phone]	[Pain point, value perception, pricing sensitivity]

5. Product & Technology

Core Features Currently Built

Feature	Status	Description
6-Checkpoint Governance Engine	Live	Multi-layer pre-execution decision validation with independent veto authority per checkpoint
Monte Carlo Validation	Live	10,000 simulation paths per decision; blocks trades with negative expected return
Adaptive Coherence Gate	Live	Dynamically calibrated signal agreement scoring based on market regime severity
Edge Confirmation Window (ECW)	Live	Requires 3 consecutive cycles of confirmed statistical edge before execution
Black Swan Detection	Live	Real-time tail risk monitoring with automatic exposure reduction (4 severity levels)
Decision Contradiction Index (DCI)	Live	Measures internal signal divergence (0–100); high DCI mandates HOLD
Post-Quantum Cryptography	Live	NIST-standardized Dilithium-3 signatures + Kyber-768 key encapsulation
Public Verification Server	Live	Public receipt verification endpoint with zero internal data exposure
Shadow Portfolio Engine	Live	670,000+ counterfactual events; validates veto accuracy (100% accuracy on 50 validated)
Non-Markovian Memory Kernel	Live	Behavioral pattern detection beyond recency bias
Multi-AI Orchestration	Live	Gemini 2.5 Flash + GPT-4o + Claude Sonnet 4 with automatic failover
Investor Dashboard	Live	19 widgets with real-time metrics, dual win rate framework, regime detection
Institutional Website	Live	Public landing at www.omnixquantum.net with live market data integration
Interactive Governance Demos	Live	Credit/Lending + Insurance underwriting demos showing multi-vertical applicability

What Is Live vs What Is Planned

Component	Status	Timeline
Governance Engine (Digital Assets)	LIVE	Since Nov 2025

Component	Status	Timeline
PQC Decision Signing	LIVE	Since Nov 2025
Public Verification System	LIVE	Since Feb 2026
Enterprise API (Risk Guardian)	Planned	Q2 2026
White-Label SDK	Planned	Q3 2026
Supply Chain Domain Adapter	Planned	Year 2–3
Credit/Lending Domain Adapter	Planned	Year 2–3
Insurance Domain Adapter	Planned	Year 3+
Robotics / Autonomous Systems Adapter	Planned	Year 3+

Product Roadmap (Next 6–12 Months)

- **Q1 2026:** Complete track record validation. Finalize institutional documentation. Eureka GCC competition.
- **Q2 2026:** Enterprise API launch (Risk Guardian API). First enterprise pilot (prop firm or trading platform). ADGM regulatory structure initiated.
- **Q3 2026:** Public license model launch. White-label SDK for platform integrations. Second and third enterprise clients.
- **Q4 2026:** Series A readiness with validated revenue metrics. Begin multi-vertical domain adapter development.

Technology Stack

Layer	Technology
Language	Python 3.11 (core engine)
Frontend	React 18 + TypeScript + Tailwind CSS + Vite
Database	PostgreSQL (42+ tables, 90% FK coverage)
Cache	Redis (state management, rate limiting)
AI Models	Google Gemini 2.5 Flash, OpenAI GPT-4o, Anthropic Claude Sonnet 4
Cryptography	CRYSTALS-Dilithium-3 (ML-DSA-65), CRYSTALS-Kyber-768 (ML-KEM-768)
Infrastructure	Railway (production 24/7), Replit (development)
Verification	aiohttp async server (port 8000), SHA-256 hash chain
Architecture	Hexagonal (ports and adapters), 27 Architecture Decision Records

Key Technical or Execution Risks

- **Key-person dependency:** Solo founder. Mitigated by documented hexagonal architecture (27 ADRs, 2–3 week onboarding target). First 3 hires planned within Month 1–4 post-funding.

- **AI model dependency:** Reliance on third-party AI providers. Mitigated by multi-provider failover chain (Gemini → GPT-4o → Claude).
- **Market timing:** Enterprise sales cycles are 3–6 months. Conservative pipeline assumptions account for this.

Demo Links: Public verification: omnibotgenesis-production.up.railway.app/verify | Website: www.omnixquantum.net | Credit governance demo: www.omnixquantum.net/governance-demo | Insurance governance demo: www.omnixquantum.net/governance-demo-insurance

6. Competitive Landscape & Differentiation

OMNIX operates in the decision governance infrastructure space, which is a new category. The closest comparisons come from adjacent markets:

Competitor	Category	Decision Checkpoints	PQC Security	Multi-Vertical	Pricing
3Commas	Trading Platform	1 (basic SL/TP)	No	No	\$49–\$79/mo
Gauntlet Networks	DeFi Risk	2–3	No	DeFi only	Enterprise
Chainalysis	Compliance	0 (monitoring only)	No	Compliance only	\$100K+/yr
Internal Quant Teams	Prop Funds	2–4	Rare	No	\$1M+/yr team cost
OMNIX	Decision Governance	6 independent	Yes (NIST)	Yes (6+ verticals)	\$15K–\$35K/mo

Unique Selling Proposition (USP)

OMNIX is the first domain-agnostic Decision Governance Infrastructure with production-integrated post-quantum cryptography, 6 independent validation checkpoints, and a publicly verifiable decision audit trail. No competitor offers this combination.

Why We Win

Today:

- **Fail-closed by default:** Every other system defaults to execute. OMNIX defaults to protect.
- **Post-quantum security in production:** No competitor has PQC-signed decision receipts.
- **Full audit trail:** 16,000+ governance receipts, publicly verifiable, zero information leakage.
- **Domain-agnostic architecture:** Same engine validates trading, credit, insurance, and robotics/autonomous systems decisions.

At Scale:

- **Network effects:** More decisions governed = better calibration. Shadow Portfolio learns from every veto.
- **Multi-vertical expansion:** One architecture, multiple verticals (trading, supply chain, lending, insurance, energy, robotics). Each new domain increases TAM without rebuilding core.
- **Regulatory moat:** MiCA, ADGM, and emerging compliance requirements create demand for auditable decision governance.
- **Switching costs:** OMNIX embeds into execution flow — high switching cost once integrated.

7. Business Model & Unit Economics

Revenue Model

OMNIX operates a dual revenue model: B2B Decision Governance Licensing (80% of revenue) and B2C SaaS (20%, post-enterprise validation).

Product	Pricing	Target Customer
Risk Guardian API	\$15K–\$35K/month	Prop firms, trading platforms
White-Label Engine	\$100K+ setup + \$20K/month	Exchanges, brokers
Per-Validation API	\$0.01–\$0.05/call	High-volume platforms
B2C Pro	\$149/month	Advanced independent traders
B2C Advanced	\$499/month	Semi-professional traders, API access

Pricing Strategy

Pricing is positioned below the cost of building proprietary risk governance internally (\$1M+ annually for a quant team) while delivering institutional-grade protection. Enterprise pilots begin at discounted rates (\$10K–\$15K/month) with conversion to full pricing after validation period.

"Institutions pay for what blocks bad decisions, not for alpha. License-based revenue. No tokens. No performance fees."

Customer Acquisition Channels

- **Direct outreach:** 50 targeted contacts/month via LinkedIn + email (targeting 15 meetings/month)
- **ADGM Innovation Hub:** Regulatory community + startup programs (5–10 warm intros/quarter)
- **Hub71:** Applied — pending response (3–5 qualified leads/quarter if accepted)
- **Industry events:** TOKEN2049 Dubai, FinTech Abu Dhabi, GITEX (10+ contacts/event)
- **Eureka GCC:** Competition exposure + judge network (immediate pipeline)

Unit Economics (Estimated)

Metric	Value	Basis
CAC (Enterprise)	\$5,000–\$10,000	Direct sales, 3–6 month cycle, founder-led
LTV (Enterprise)	\$180K–\$420K	\$15K–\$35K/month × 12-month minimum contract
LTV/CAC Ratio	18x–84x	Strong unit economics at enterprise tier
CAC (B2C SaaS)	\$50–\$150	Content marketing + referral programs
LTV (B2C SaaS)	\$1,788–\$5,988	\$149–\$499/month × 12-month avg retention

Metric	Value	Basis
Gross Margin	60–70%	SaaS infrastructure model
Break-Even	18–24 months	Conservative estimate

8. Traction & Key Metrics

OMNIX has been running in production 24/7 since November 2025. The following metrics represent real system telemetry, not projections:

Metric	Value	Significance
Production Uptime	24/7 since Nov 2025	System operational for 4+ months continuously
Evaluation Cycles	670,000+	Governance engine processing decisions in real-time
PQC-Signed Receipts	16,000+	Every decision signed with Dilithium-3 (100% coverage)
Capital Preserved	98.5%	During period when BTC dropped 7.37%
Shadow Trade Events	670,000+	Counterfactual analysis of vetoed decisions
Veto Accuracy	100%	50 validated shadow outcomes — all vetos were correct
Decision Latency	<120ms	Real-time governance validation
Database Tables	42+	90% foreign key coverage — institutional-grade data model
Architecture Decisions	27 ADRs	Documented engineering discipline
Dashboard Widgets	19/19	Full operational visibility

Competition & Recognition

- **Eureka GCC 2026:** Semifinalist — competing in Abu Dhabi startup competition
- **Public verification:** Live at omnibotgenesis-production.up.railway.app/verify
- **Institutional website:** Live at www.omnixquantum.net with real-time market data

Key Learning Milestones

Date	Milestone	Impact
Nov 2025	PQC implemented (Dilithium-3 + Kyber-768)	Quantum-resistant decision signing operational
Nov–Dec 2025	Learning Baseline (119 trades)	System calibration, threshold discovery
Jan 15, 2026	Official Track Record Day 1	Clean metrics from calibrated parameters
Jan 21, 2026	Edge Confirmation Window + DCI	Capital patience + contradiction detection
Jan 28, 2026	Institutional Website launched	Public credibility and demo capability

Date	Milestone	Impact
Feb 15, 2026	Multi-Vertical Demos (Credit + Insurance)	Domain-agnostic architecture demonstrated
Feb 21, 2026	Public Verification Server	Transparent governance receipts
Feb 23, 2026	Railway cost optimization (77% reduction)	Operational efficiency for scaling

Note: All metrics are from internal dataset, not externally audited. Evaluation cycles and receipts represent governance engine activity, not trading performance.

9. Go-To-Market Strategy

Customer Acquisition Strategy

OMNIX follows an enterprise-first go-to-market strategy, targeting institutional buyers in the ADGM/DIFC ecosystem where regulatory urgency (MiCA 2025+) creates immediate demand for decision governance infrastructure.

Stage	Action	Timeline	Conv. Rate
Awareness	50 direct outreaches/month + event networking	Ongoing	—
Assessment	Free governance health check for their system	Week 1–2	30%
Shadow Mode	OMNIX runs alongside their system (no execution)	2–4 weeks	60%
Advisory Mode	OMNIX provides pre-execution recommendations	2–4 weeks	75%
Enforcement	Full governance integration with veto authority	Ongoing	80%
Paid License	Monthly enterprise license	Month 3–6	—

Sales Cycle

Enterprise sales cycle: 3–6 months from first contact to paid license. The progressive onboarding model (Shadow → Advisory → Enforcement) reduces adoption risk and builds trust before requiring financial commitment. Founder-led sales during pre-seed phase.

Key Partnerships (Target)

Partner Type	Target	Status
Regulatory Ecosystem	ADGM (Abu Dhabi)	Target licensing ecosystem
Accelerator	Hub71	Applied — pending response
Future Expansion	DIFC (Dubai)	Roadmap
Technology	Cloud infrastructure (Railway)	Active
AI Providers	Google, OpenAI, Anthropic	Active (multi-provider)

Scaling Strategy Across Markets

Phase	Focus	Timeline
Phase 1	Pilot validation — 3 enterprise clients (digital assets)	Months 1–6
Phase 2	Regional expansion (ADGM/DIFC/MENA) + platform partnerships	Months 6–12
Phase 3	Multi-vertical expansion — supply chain, lending, robotics/autonomous systems	Months 12–24
Phase 4	Global scaling (EU MiCA, Asia) + insurance, energy	Months 24–36

10. Team & Advisors

Founder

Harold Nunes — Founder & CEO

- Product architecture, risk logic design, and infrastructure development
- Built OMNIX end-to-end: from concept to production system running 24/7
- AI-augmented development methodology: one person with AI achieves the output of a 5-person team
- Deep domain expertise in algorithmic trading risk management, decision systems, and cryptographic security
- Designed the 6-checkpoint governance architecture, post-quantum integration, and multi-vertical domain adapter pattern

Key Hires Planned (Post-Funding)

Role	Timeline	Purpose
Senior Backend Engineer	Month 1–2	Enterprise API development, system scaling
DevOps / Infrastructure	Month 2–3	Production reliability, deployment automation
Business Development	Month 3–4	Enterprise sales, ADGM ecosystem relationships

Key-Person Risk Mitigation

The current key-person risk is acknowledged and actively mitigated:

- **Documented architecture:** 27 Architecture Decision Records (ADRs), hexagonal architecture with clear module boundaries
- **Onboarding target:** 2–3 week onboarding for new engineers using existing documentation
- **Dependency reduction:** First 3 hires reduce founder dependency from 100% to ~30% by Month 4
- **IP protection:** IP assignment to company entity, key-person insurance, operational runbooks by Month 6

Advisors

Advisory board is being assembled. The founder has worked with contract developers and infrastructure consultants during the development phase. Formal advisory relationships with ADGM ecosystem participants and fintech domain experts are targeted post-funding.

Why This Team

OMNIX was built by a founder who understands both the technical complexity of multi-layer governance systems and the institutional requirements of regulated financial markets. The AI-augmented development model has enabled solo construction of a system that would typically require a 5-person engineering team — evidenced by 42+ database tables, 27 ADRs, 670,000+ evaluation cycles, and 4+ months of continuous production operation.

Post-funding, the first hires are designed to eliminate key-person risk and accelerate enterprise sales.

11. Financials & Fundraising

3-Year Financial Projections

	Year 1	Year 2	Year 3
Enterprise Clients	3 pilots	5–8 licenses	15–30 licenses
B2C SaaS Users	50–100	500–1,000	2,000–5,000
Revenue	\$200K–\$400K	\$800K–\$1.2M	\$2M+
Gross Margin	60–70%	65–75%	70–80%
Key Driver	Enterprise pilots	License expansion	Multi-vertical + SaaS scale

Key Assumptions

- 3 enterprise pilots secured within 6 months via ADGM/DIFC ecosystem
- Average enterprise contract: \$25K/month (mid-range of \$15K–\$35K pricing)
- 12-month minimum contract duration with 80% renewal rate
- B2C SaaS launched post-enterprise validation (Month 6–9)
- Infrastructure costs scale linearly with client count at 60–70% gross margin
- MiCA compliance urgency drives accelerated adoption in EU markets (Year 2+)

Fundraising

Item	Details
Raising	\$500,000 USD
Equity	16.7%
Pre-Money Valuation	\$2.5M–\$3M
Funds Raised to Date	\$0 (bootstrapped)
Instrument	Equity (pre-seed round)

Valuation Justification

- **Working product in production:** 4+ months running 24/7 with real market data
- **Real validation data:** 670,000+ evaluation cycles, 16,000+ PQC-signed receipts
- **Defensible IP:** 6-checkpoint architecture + Shadow Portfolio engine + PQC integration
- **Strategic timing:** MiCA + ADGM convergence creating urgent demand for governance infrastructure
- **Comparable:** Chainalysis raised at \$4M pre-money at similar stage

Use of Funds

Category	Allocation	Purpose
Strategy & Risk Engine	35%	Algorithm refinement, Shadow Portfolio expansion, enterprise API
Dubai/ADGM Legal & Regulatory	25%	Company formation, regulatory structure, compliance
Enterprise Infrastructure	20%	API for prop firms, security certifications, scaling
Team & Operations	15%	2–3 key hires eliminating key-person risk (Month 1–4)
Reserve	5%	Contingency

12. Intellectual Property & Regulatory

Intellectual Property

OMNIX's intellectual property is embedded in its proprietary architecture and algorithms, not in traditional patents. The key defensible assets are:

IP Asset	Description	Protection
6-Checkpoint Governance Engine	Multi-layer pre-execution decision validation architecture	Trade secret + documented ADRs
Adaptive Coherence Gate	Dynamic threshold calibration based on market regime	Proprietary algorithm
Edge Confirmation Window	Statistical edge persistence validation (3-cycle)	Proprietary methodology
Decision Contradiction Index	Internal signal divergence measurement (0–100)	Proprietary metric
Shadow Portfolio Engine	670,000+ counterfactual events for veto validation	Proprietary dataset + methodology
Non-Markovian Memory Kernel	Behavioral pattern detection beyond recency bias	Proprietary algorithm
PQC Integration Architecture	Production-integrated post-quantum decision signing	First-mover advantage

Patent filing strategy will be evaluated post-funding with legal counsel in ADGM jurisdiction. Current protection relies on trade secrets, first-mover advantage, and architectural complexity.

Regulatory Framework

Framework	Status	Relevance
ADGM (Abu Dhabi)	Target ecosystem	Company formation + licensing structure
MiCA (EU)	Creating demand	Decision governance documentation required for compliance
GDPR	Controls aligned	Data protection and PII handling
SOC 2 Principles	Aligned	Access control, audit logging, encryption
MiFID II	Aligned	Decision auditability and traceability
DORA	Aligned	Operational resilience and error isolation

Compliance Risks & Mitigation

- **Regulatory classification:** OMNIX is infrastructure (not a financial product or fund). No asset custody, no investment advice, no token issuance.
- **Data residency:** PostgreSQL database with configurable deployment region. ADGM/DIFC data handling requirements addressed through cloud infrastructure selection.
- **Cryptographic compliance:** NIST-standardized algorithms (Dilithium-3, Kyber-768). No proprietary or unvetted cryptographic implementations.

13. Risks & Challenges

Top 3 Business Risks

1. Key-Person Risk (HIGH)

OMNIX is currently built and operated by a solo founder. This creates dependency risk for investors and operational continuity.

Mitigation: Documented architecture (27 ADRs), hexagonal module design with clear boundaries, 2–3 week onboarding target. First 3 hires (Month 1–4) reduce dependency from 100% to ~30%. IP assignment, key-person insurance, and operational runbooks by Month 6.

2. Enterprise Sales Cycle Risk (MEDIUM)

Enterprise sales cycles are typically 3–6 months. Delayed pilot agreements could impact runway and milestone timelines.

Mitigation: Progressive onboarding model (Shadow → Advisory → Enforcement) reduces adoption friction. ADGM/DIFC ecosystem provides warm introductions. MiCA compliance urgency accelerates decision-making. \$500K runway provides 18–24 month operating buffer.

3. Market Timing Risk (MEDIUM)

Crypto market volatility could reduce institutional appetite for digital asset infrastructure during bear markets.

Mitigation: OMNIX is positioned as decision governance infrastructure, not a crypto product. The multi-vertical architecture (supply chain, lending, insurance, robotics/autonomous systems) reduces dependency on any single market. Bear markets actually increase demand for risk governance.

Execution Challenges

- **Scaling from solo to team:** First hires must maintain code quality and architectural discipline. Hexagonal architecture and ADR documentation mitigate this.
- **Enterprise integration complexity:** Each client requires customized onboarding. Domain Adapter pattern standardizes this process.
- **AI provider reliability:** Multi-provider failover chain (Gemini → GPT-4o → Claude) ensures no single point of failure.

Market & Regulatory Risks

- **Regulatory changes:** ADGM/DIFC regulations could shift. OMNIX's compliance-aligned architecture makes adaptation straightforward.
- **Competition from incumbents:** Large exchanges could build in-house governance. OMNIX's 4-month head start, PQC integration, and multi-vertical design (6+ verticals including robotics) create defensible position.
- **AI regulation:** Emerging AI governance regulations (EU AI Act) could affect multi-AI orchestration. OMNIX's transparent decision audit trail is aligned with explainability requirements.

14. Appendix

A. Architecture Overview

Decision Engine
+-- Signal Detection (EMA Regime, HMM, Kalman, Non-Markovian, Kelly)
+-- 6-Checkpoint Governance Flow:
1. Monte Carlo VETO (10,000 simulations)
2. RMS VETO (VaR95, drawdown limits)
3. Adaptive Coherence Gate (dynamic thresholds)
4. Edge Confirmation Window (3-cycle persistence)
5. Weighted Scoring (5 inputs, 100 points)
6. Final Decision (EXECUTE / HOLD / BLOCK)
+-- Post-Decision:
+-- Dilithium-3 PQC Signature
+-- SHA-256 Hash Chain
+-- PostgreSQL Persistence
+-- Public Verification Endpoint

B. Scoring Logic

Input	Weight	Max Points
EMA Regime Signal	40%	40 pts
HMM Regime Detection	25%	25 pts
Kalman Filter	15%	15 pts
Non-Markovian Memory	15%	15 pts
Kelly Criterion	10%	10 pts (sizing only)

Separate Veto/Penalty layer (Monte Carlo, Black Swan, Sentiment, Quantum Momentum) applies only penalties — it cannot add positive score.

C. Decision Receipt Example

```
{  
  
  "receipt_id": "RCP-2026022115...",  
  
  "timestamp": "2026-02-21T23:55:39Z",  
  
  "asset": "BTC/USD",  
  
  "decision": "HOLD",  
  
  "content_hash": "sha256:a7f3e2...",  
  
  "prev_hash": "sha256:8b4c1d... (truncated)",  
  
  "signature_algorithm": "Dilithium-3 (ML-DSA-65)",  
  
  "governance_gates": [  
  
    "MC_VETO", "RMS_CHECK", "COHERENCE_GATE",  
  
    "ECW_GATE", "SCORING", "FINAL_DECISION"  
  
  ]  
  
}
```

D. Links & Resources

- **Website:** www.omnixquantum.net
- **Public Verification:** omnibotgenesis-production.up.railway.app/verify
- **Credit Governance Demo:** www.omnixquantum.net/governance-demo
- **Insurance Governance Demo:** www.omnixquantum.net/governance-demo-insurance
- **LinkedIn:** linkedin.com/in/harold-nunes-21bb65285
- **Contact:** contacto@omnixquantum.net

OMNIX — Decision Governance Infrastructure for Automated Systems
"When systems can act, but choose discipline."

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All metrics from internal dataset, not externally audited. Past performance does not guarantee future results.